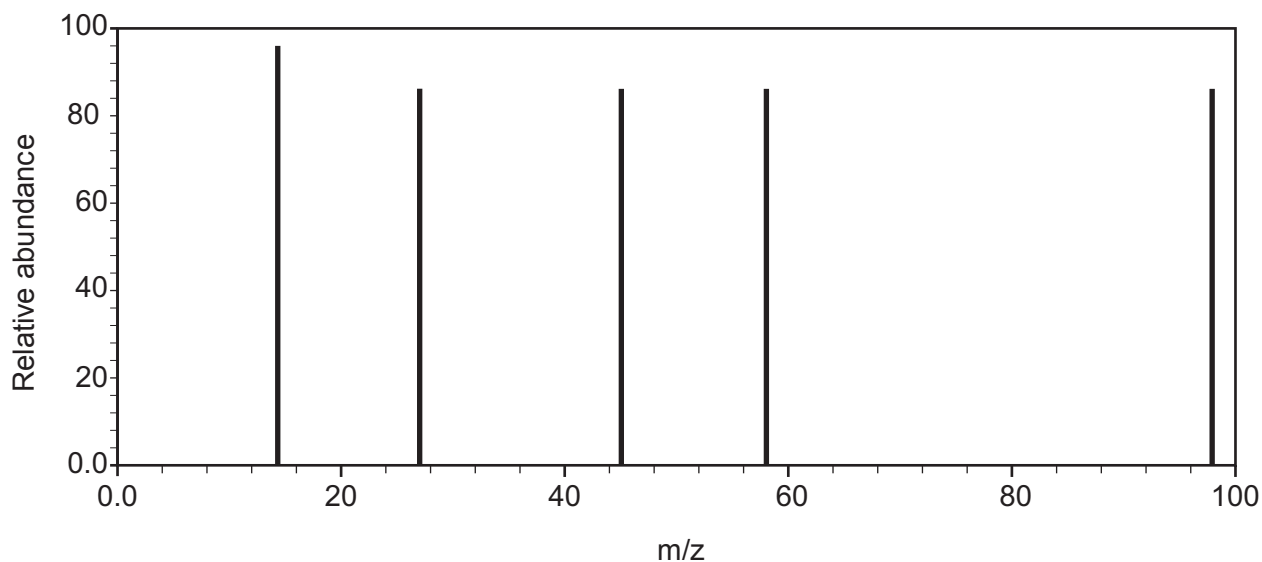


12. Chemists are investigating an unknown compound **X**. They obtain information from a variety of sources.

Compound **X** contains 61.2% carbon, 6.1% hydrogen and 32.7% oxygen by mass. A simplified mass spectrum of compound **X** is shown below.



When solid sodium carbonate is added to an aqueous solution of **X** effervescence is observed.

There are 5 peaks in the ^{13}C NMR spectrum of **X**.

1 mol of **X** reacts completely with 320 g of bromine in the dark.



Use all the data given to find the structure of compound **X**. Explain what information can be found from each piece of data. [10]

Examiner
only

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END OF PAPER



Surname	Centre Number	Candidate Number
Other Names		2


GCE AS/A LEVEL – NEW

2410U20-1



S17-2410U20-1

CHEMISTRY – AS unit 2
Energy, Rate and Chemistry of Carbon Compounds

FRIDAY, 9 JUNE 2017 – AFTERNOON

1 hour 30 minutes

For Examiner's use only		
Question	Maximum Mark	Mark Awarded
Section A 1. to 6.	10	
Section B 7.	14	
8.	16	
9.	13	
10.	14	
11.	13	
Total	80	

ADDITIONAL MATERIALS

In addition to this examination paper, you will need a:

- calculator;
- **Data Booklet** supplied by WJEC.

INSTRUCTIONS TO CANDIDATES

Use black ink or black ball-point pen. Do not use gel pen or correction fluid.

Write your name, centre number and candidate number in the spaces at the top of this page.

Section A Answer **all** questions in the spaces provided.

Section B Answer **all** questions in the spaces provided.

 Candidates are advised to allocate their time appropriately between **Section A (10 marks)** and **Section B (70 marks)**.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets at the end of each question or part-question.

The maximum mark for this paper is 80.

Your answers must be relevant and must make full use of the information given to be awarded full marks for a question.

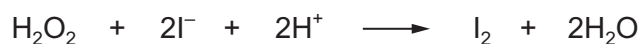
 The assessment of the quality of extended response (QER) will take place in **Q.8(b)**.

If you run out of space, use the additional page(s) at the back of the booklet, taking care to number the question(s) correctly.



JUN172410U20101

12. (a) Iodide ions can be oxidised to iodine by reaction with acidified hydrogen peroxide.



The rate of reaction can be followed in a clock reaction by the appearance of a blue-black colour.

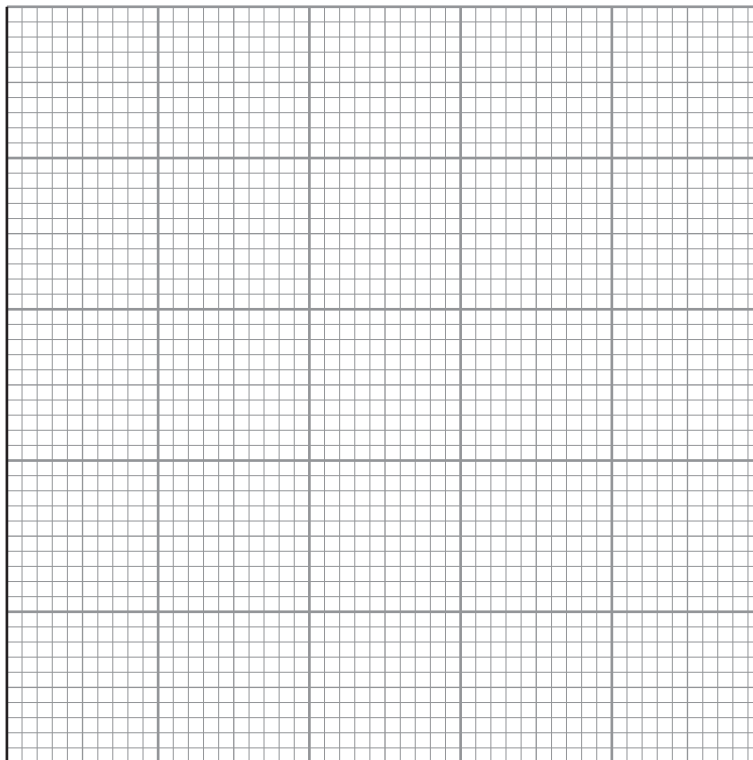
An experiment was carried out to determine the effect on the rate of reaction of varying the concentration of iodide ions. All other volumes and concentrations were kept constant. The results are shown in the table.

Concentration I^- / mol dm^{-3}	Time for appearance of blue-black colour / s	Rate / $\text{s}^{-1} \times 1000$
0.1	56	18
0.2	20	
0.3	18	
0.4	12	
0.5	10	

- (i) Use $\text{rate} = \frac{1000}{\text{time}}$ to calculate the rate for each experiment and complete the table. [1]



- (ii) On the axes below, plot the concentration of I^- against rate and draw a suitable line. [3]



- (iii) From the graph deduce the relationship between concentration of I^- and rate of reaction. [1]

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- (iv) **Use the graph** to calculate the time the reaction would take to turn blue-black using a 0.15 mol dm^{-3} solution of I^- . Show clearly how you obtained your answer. [2]

Time = s



- (v) For each experiment the rate was calculated using the time taken to produce excess iodine. Explain why this is only an approximation for the rate **as the reaction proceeds**. [2]

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- (b) (i) Draw a Boltzmann energy distribution curve. Label the axes. [2]

- (ii) Use this energy distribution curve to explain how catalysts affect the rate of a reaction. [2]

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12. (a) **Y** is a halogenocompound in which each molecule contains one atom of chlorine, bromine or iodine.

(i) Describe a chemical test to determine which halogen is present in **Y**. [3]

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(ii) **Y** contains four carbon atoms in each molecule. The percentage by mass of halogen present in **Y** is less than 40%.

Identify **Y**. Explain how you reached your conclusion. [2]

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- (b) (i) Halogenoalkanes react with aqueous sodium hydroxide.

Draw the mechanism to show the reaction of 1-chloropropane with aqueous sodium hydroxide.

You should include all charges, partial charges and lone pairs, and curly arrows to show electron movement. [4]

- (ii) Name the type of reaction shown in part (i). [1]

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- (c) Halogenoalkanes can also take part in an elimination reaction.

2-Chloropentane undergoes elimination in a similar way to 1-chloropropane.

- (i) Give the reagent and conditions needed for 2-chloropentane to undergo elimination. [1]

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- (ii) When 2-chloropentane undergoes elimination, two structural isomers are formed.

Give the structures of these **two** isomers. [2]

